RAG Learning

2.2 Data Presentation & Interpretation

Mathematics B (MEI) — OCR B — A-Level (H640) — Statistics — spec refs D1–D14

What this topic covers. Types of data and the diagrams that suit them, measures of central tendency and spread, variance and standard deviation, identifying outliers, cleaning data, and interpreting scatter diagrams and correlation.

1. Types of data

Type	Meaning	Example
Categorical	Labels, not numbers	Eye colour
Discrete	Numerical, taking separate values	Number of cars
Continuous	Numerical, any value in a range	Height
Ranked	Ordered positions rather than values	Finishing places

2. Diagrams

Diagram	Suited to	Note
Bar chart	Categorical	Bars separated; height is frequency
Vertical line chart	Discrete numerical	Lines rather than bars
Dot plot	Small data sets	Stacks of dots represent frequency

Histogram	Continuous, grouped	**Area** is proportional to frequency
Frequency chart	Grouped, equal widths	Equal-width bars; vertical axis is **frequency**
Pie chart	Proportions of a whole	Comparative pie charts have **area** proportional to frequency
Stem-and-leaf	Small sets	Retains the original values; needs a key
Box plot	Comparing distributions	Shows median, quartiles and extremes
Cumulative frequency	Estimating median and quartiles	Plot against upper class boundaries

The **histogram versus frequency chart** distinction is specific to MEI and is examined. A histogram may have unequal class widths and uses frequency density; a frequency chart has equal widths and plots frequency itself.

Histograms

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}, \qquad \text{frequency} = \text{density} \times \text{width}$$

Worked example. A histogram bar has width 5 and frequency density 3. Its frequency is $5 \times 3 = 15$.

Comparative pie charts

When two pie charts represent data sets of different sizes, the **areas** are made proportional to the total frequencies. Since area $\propto r^2$, the radii are proportional to $\sqrt{\text{frequency}}$.

Worked example. Two data sets have totals 50 and 200. The ratio of radii is $\sqrt{50} : \sqrt{200} = 1 : 2$, not $1 : 4$.

3. Measures of central tendency

Measure	Definition	When to prefer it
Mean	$\bar{x} = \frac{\sum x}{n}$	Symmetric data with no outliers
Median	Middle value in order	Skewed data, or when outliers are present
Mode	Most frequent value	Categorical data
Midrange	$\frac{\min + \max}{2}$	Quick summary; very sensitive to extremes

A **weighted mean** is appropriate when combining groups of different sizes — for example using populations as weights.

MEI states that the focus is on **interpretation rather than calculation**. Expect to be asked which average is appropriate and why, more often than to compute one by hand.

4. Measures of spread

Measure	Definition
Range	Maximum − minimum
Interquartile range	$Q_3 - Q_1$
Percentiles	Values below which a given percentage falls
Variance	$s^2 = \frac{S_{xx}}{n-1}$
Standard deviation	$s = \sqrt{s^2}$

where

$$S_{xx} = \sum (x_i - \bar{x})^2$$

The **sample** variance divides by $n - 1$, not n. This is what makes it an unbiased estimator of the population variance. Dividing by n is the standard error and produces a value that is slightly too small.

Worked example. A data set of 10 values has $\sum x = 200$ and $S_{xx} = 360$.

$$\bar{x} = \frac{200}{10} = 20, \qquad s^2 = \frac{360}{9} = 40, \qquad s = \sqrt{40} \approx 6.32$$

Standard deviation carries the same units as the data; variance carries their square.

Effect of transforming the data

Transformation	Mean	Standard deviation
Add a constant c	Increases by c	**Unchanged**
Multiply by k	Multiplied by k	Multiplied by $\lvert k \rvert$

Adding a constant shifts every value equally, so the *spread about the mean* is untouched. This is worth stating explicitly when justifying the answer.

5. Outliers

An **outlier** is an item inconsistent with the rest of the data. MEI gives two accepted criteria (spec ref D13): a value at least **2 standard deviations** from the mean, *or* at least **1.5 × IQR** beyond the nearer quartile.

Worked example. Mean 50, standard deviation 4. By the first criterion, outliers are values below $50 - 8 = 42$ or above $50 + 8 = 58$.

Worked example. $Q_1 = 20$, $Q_3 = 32$, so $\mathrm{IQR} = 12$ and $1.5 \times \mathrm{IQR} = 18$. Outliers are values below $20 - 18 = 2$ or above $32 + 18 = 50$.

State which criterion you are using. The two do not always agree, and a question that gives you quartiles is signalling the IQR rule while one giving a standard deviation signals the other.

6. Cleaning data

Cleaning covers missing values, errors and outliers (D14).

- An impossible value — an adult height recorded as 1750 cm — is a data-entry error and should be corrected or removed, with the reason stated.
- A genuine extreme value should generally be **kept**: it is real information, and discarding it distorts the conclusions.
- Missing data should be acknowledged, not silently ignored.

Removing an outlier requires **justification**. "It looked wrong" is not a reason; "it is physically impossible for an adult height" is.

7. Bivariate data and correlation

A **scatter diagram** displays paired observations. Correlation measures how closely the points lie to a straight line (H10).

Pattern	Description
Points rising left to right	Positive correlation
Points falling left to right	Negative correlation
No pattern	Little or no correlation

Correlation does not imply causation. Ice cream sales and drownings correlate strongly, but neither causes the other — hot weather drives both. Always consider a possible third variable.

Regression lines

A line or curve of best fit lets you estimate one variable from the other.

- **Interpolation** — estimating within the range of the data. Generally reliable.
- **Extrapolation** — estimating beyond it. Not justified, since the relationship may not continue.

What this specification does and does not require. You are expected to *interpret* a regression line or other best-fit model, including judging interpolation and extrapolation — but **calculating** the equation of a regression line is excluded. Likewise, you use a *given* correlation coefficient to make an inference; calculating one, and knowing the names of particular coefficients, are both excluded. A **rank** correlation coefficient measures correlation between the ranks rather than the raw values.

8. Sample size and shape

Diagrams from unbiased samples become more representative of the underlying theoretical distribution as the sample grows (D5). Tossing a fair coin, the proportion of heads moves steadily closer to 50% as the number of tosses increases.

9. Common pitfalls

- Dividing by n instead of $n - 1$ for the sample variance.
- Treating a histogram's bar height as the frequency.
- Making the radius, rather than the area, proportional to frequency in comparative pie charts.
- Claiming causation from correlation.
- Extrapolating beyond the data without noting that it is unjustified.
- Attempting to calculate a regression line or correlation coefficient, which is off-specification.
- Removing an outlier without justification.

- Giving standard deviation in squared units.